

Sarthi Healthcare Platform - Technical Appendices

Implementation Details & Operational Procedures

Appendix A: GCP Infrastructure Requirements & Setup Guide

A.1 GCP Organization Structure & Prerequisites

Organization Hierarchy

```
yaml
Organization: sarthi-healthcare.com
  Folder: Production
    Project: sarthi-prod-core
    Project: sarthi-prod-data
    Project: sarthi-prod-ai
  Folder: Non-Production
    Project: sarthi-staging
    Project: sarthi-dev
  Folder: Shared
    Project: sarthi-security
    Project: sarthi-networking
    Project: sarthi-billing
```

Initial Setup Requirements

```
bash
```

Required APIs to Enable per Project

gcloud services **enable** compute.googleapis.com
gcloud services **enable** container.googleapis.com
gcloud services **enable** cloudrun.googleapis.com
gcloud services **enable** cloudfunctions.googleapis.com
gcloud services **enable** cloudbuild.googleapis.com
gcloud services **enable** cloudkms.googleapis.com
gcloud services **enable** secretmanager.googleapis.com
gcloud services **enable** healthcare.googleapis.com
gcloud services **enable** sqladmin.googleapis.com
gcloud services **enable** redis.googleapis.com
gcloud services **enable** storage.googleapis.com
gcloud services **enable** bigquery.googleapis.com
gcloud services **enable** pubsub.googleapis.com
gcloud services **enable** cloudtasks.googleapis.com
gcloud services **enable** cloudscheduler.googleapis.com
gcloud services **enable** monitoring.googleapis.com
gcloud services **enable** logging.googleapis.com
gcloud services **enable** cloudtrace.googleapis.com
gcloud services **enable** firebase.googleapis.com
gcloud services **enable** firestore.googleapis.com
gcloud services **enable** firebasehosting.googleapis.com
gcloud services **enable** documentai.googleapis.com
gcloud services **enable** aiplatform.googleapis.com
gcloud services **enable** translate.googleapis.com
gcloud services **enable** speech.googleapis.com
gcloud services **enable** texttospeech.googleapis.com

IAM Roles & Service Accounts

yaml

Service_Accounts:

- **name:** sarthi-api-gateway@sarthi-prod-core.iam

roles:

- roles/cloudsql.client
- roles/datastore.user
- roles/pubsub.publisher
- roles/cloudtasks.enqueueuer
- roles/secretmanager.secretAccessor

- **name:** sarthi-ai-agent@sarthi-prod-ai.iam

roles:

- roles/aiplatform.user
- roles/documentai.apiUser
- roles/healthcare.fhirResourceReader
- roles/healthcare.fhirResourceWriter

- **name:** sarthi-data-pipeline@sarthi-prod-data.iam

roles:

- roles/bigquery.dataEditor
- roles/pubsub.subscriber
- roles/storage.objectViewer

- **name:** firebase-hosting@sarthi-prod-core.iam

roles:

- roles/firebasehosting.admin
- roles/firebaseauth.admin
- roles/datastore.user

A.2 Firebase UI Layer Configuration

Firebase Project Setup

yaml

Firestore_Configuration:

project_id: sarthi-prod-core

hosting:

site: sarthi-healthcare

public_directory: out # Next.js static export

regions:

- us-central1
- us-east1

custom_domains:

- app.sarthi.healthcare
- www.sarthi.healthcare

authentication:

providers:

- email/password
- google
- microsoft # For healthcare org SSO
- saml # Enterprise SSO

multi_tenancy:

enabled: true

tenant_separation: true

custom_domains: true

security_rules:

session_duration: 8h

mfa_enforcement: required

password_policy:

- min_length: 12
- require_uppercase: true
- require_numeric: true
- require_special: true

firestore:

mode: Native

regions:

- nam5 (United States multi-region)

collections:

- users
- tenants
- sessions

- audit_logs
- preferences
- notifications

security_rules: |

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {
    // Multi-tenant data isolation
    match /tenants/{tenantId}/{document=**} {
      allow read, write: if request.auth != null
        && request.auth.token.tenant_id == tenantId;
    }

    // User profile access
    match /users/{userId} {
      allow read, write: if request.auth != null
        && request.auth.uid == userId;
    }

    // Audit logs - write only
    match /audit_logs/{document=**} {
      allow write: if request.auth != null;
      allow read: if request.auth.token.admin == true;
    }
  }
}
```

storage:

buckets:

- **name:** sarthi-user-uploads

location: US

rules: |

```
rules_version = '2';
service firebase.storage {
  match /b/{bucket}/o {
    match /tenants/{tenantId}/{allPaths=**} {
      allow read, write: if request.auth != null
        && request.auth.token.tenant_id == tenantId
        && request.resource.size < 50 * 1024 * 1024; // 50MB limit
    }
  }
}
```

```
app_check:  
  enabled: true  
  enforcement_mode: ENFORCED  
  recaptcha_v3_site_key: ${RECAPTCHA_SITE_KEY}
```

```
performance_monitoring:  
  enabled: true  
  sampling_rate: 0.1 # 10% sampling
```

```
remote_config:  
  parameters:  
    - maintenance_mode: false  
    - feature_flags: {}  
    - api_endpoints: {}  
    - ui_config: {}
```

Firestore Hosting Configuration (firebase.json)

```
json
```

```
{
  "hosting": {
    "public": "out",
    "ignore": ["firebase.json", "**/.*", "**/node_modules/**"],
    "rewrites": [
      {
        "source": "/api/**",
        "run": {
          "serviceId": "sarathi-api-gateway",
          "region": "us-central1"
        }
      },
      {
        "source": "**",
        "destination": "/index.html"
      }
    ],
    "headers": [
      {
        "source": "**/*.@(jpg|jpeg|gif|png|svg|webp)",
        "headers": [
          {
            "key": "Cache-Control",
            "value": "public, max-age=31536000, immutable"
          }
        ]
      },
      {
        "source": "**/*.@(js|css)",
        "headers": [
          {
            "key": "Cache-Control",
            "value": "public, max-age=86400, s-maxage=31536000"
          }
        ]
      },
      {
        "source": "**",
        "headers": [
          {
            "key": "X-Frame-Options",
            "value": "SAMEORIGIN"
          }
        ],
      }
    ]
  }
}
```

```

    {
      "key": "X-Content-Type-Options",
      "value": "nosniff"
    },
    {
      "key": "Strict-Transport-Security",
      "value": "max-age=31536000; includeSubDomains"
    },
    {
      "key": "Content-Security-Policy",
      "value": "default-src 'self'; script-src 'self' 'unsafe-inline' https://apis.google.com; style-src 'self' 'unsafe-inline'; im
    }
  ]
}
]
},
"firebase": {
  "firestore": {
    "rules": "firestore.rules",
    "indexes": "firestore.indexes.json"
  },
  "storage": {
    "rules": "storage.rules"
  },
  "emulators": {
    "auth": {
      "port": 9099
    },
    "firestore": {
      "port": 8080
    },
    "hosting": {
      "port": 5000
    },
    "storage": {
      "port": 9199
    }
  }
}
}

```

A.3 Core Infrastructure Components Setup

VPC Network Configuration

yaml

VPC_Configuration:

name: sarthi-vpc

auto_create_subnetworks: false

routing_mode: REGIONAL

subnets:

- name: sarthi-subnet-us-central1

region: us-central1

ip_range: 10.0.0.0/20

private_google_access: true

flow_logs: enabled

- name: sarthi-subnet-us-east1

region: us-east1

ip_range: 10.1.0.0/20

private_google_access: true

flow_logs: enabled

firewall_rules:

- name: deny-all-ingress

priority: 65534

direction: INGRESS

action: DENY

rules: ['all']

- name: allow-internal

priority: 1000

direction: INGRESS

action: ALLOW

source_ranges: ['10.0.0.0/8']

rules: ['all']

- name: allow-health-checks

priority: 900

direction: INGRESS

action: ALLOW

source_ranges: ['35.191.0.0/16', '130.211.0.0/22']

rules: ['tcp:80,443']

cloud_nat:

name: sarthi-nat

region: us-central1

nat_ip_allocate_option: AUTO_ONLY

source_subnetwork_ip_ranges_to_nat: ALL_SUBNETWORKS_ALL_IP_RANGES

private_service_connect:

- service: servicenetworking.googleapis.com

reserved_ip_range: 10.100.0.0/16

Cloud SQL PostgreSQL Setup

yaml

CloudSQL_Instance:

name: sarthi-main-db

database_version: POSTGRES_15

region: us-central1

tier: db-custom-8-32768 # 8 vCPUs, 32GB RAM

settings:

availability_type: REGIONAL # HA configuration

backup_configuration:

enabled: true

start_time: "02:00"

location: us

point_in_time_recovery_enabled: true

transaction_log_retention_days: 7

retained_backups: 30

ip_configuration:

ipv4_enabled: false # Private IP only

private_network: projects/sarthi-prod/global/networks/sarthi-vpc

authorized_networks: []

database_flags:

- name: max_connections

value: "500"

- name: shared_buffers

value: "8GB"

- name: effective_cache_size

value: "24GB"

- name: work_mem

value: "16MB"

- name: maintenance_work_mem

value: "2GB"

- name: random_page_cost

value: "1.1" # SSD optimization

- name: log_statement

value: "all" # Audit logging

- name: log_duration

value: "on"

- name: cloudsql.enable_pgaudit

value: "on"

insights_config:

query_insights_enabled: true

query_string_length: 1024
record_application_tags: true
record_client_address: true

databases:

- name: sarthi_production
charset: UTF8
collation: en_US.UTF8

users:

- name: sarthi_app
password: \${SECRET_MANAGER_REF}
- name: sarthi_readonly
password: \${SECRET_MANAGER_REF}

replicas:

- name: sarthi-main-db-replica-1
region: us-east1
tier: db-custom-4-16384
replica_configuration:
failover_target: true

Cloud Storage Buckets

yaml

Storage_Buckets:

- name: sarthi-patient-documents-prod

location: US

storage_class: STANDARD

uniform_bucket_level_access: true

versioning: true

lifecycle_rules:

- action: SetStorageClass

storage_class: NEARLINE

age: 90

- action: SetStorageClass

storage_class: COLDLINE

age: 365

encryption:

default_kms_key: projects/sarthi-prod/locations/us/keyRings/sarthi-kr/cryptoKeys/storage-key

retention_policy:

retention_period: 2555 # 7 years for HIPAA

- name: sarthi-ai-artifacts-prod

location: US

storage_class: STANDARD

lifecycle_rules:

- action: Delete

age: 30 # Temporary AI processing artifacts

- name: sarthi-backups-prod

location: US

storage_class: NEARLINE

versioning: true

retention_policy:

retention_period: 365

A.4 Compute Resource Specifications

Cloud Run Services

yaml

Cloud_Run_Services:

sarathi-api-gateway:

cpu: "2"

memory: 4Gi

min_instances: 3

max_instances: 100

concurrency: 250

timeout: 300s

service_account: sarathi-api-gateway@sarathi-prod-core.iam

sarathi-llm-gateway:

cpu: "4"

memory: 8Gi

min_instances: 2

max_instances: 50

concurrency: 50

timeout: 540s

service_account: sarathi-ai-agent@sarathi-prod-ai.iam

sarathi-document-processor:

cpu: "2"

memory: 4Gi

min_instances: 1

max_instances: 30

concurrency: 10

timeout: 300s

sarathi-clinical-agent:

cpu: "4"

memory: 8Gi

min_instances: 2

max_instances: 40

concurrency: 25

timeout: 540s

sarathi-billing-agent:

cpu: "2"

memory: 4Gi

min_instances: 1

max_instances: 20

concurrency: 50

timeout: 300s

Cloud Functions

yaml

Cloud_Functions:

voice-transcription-handler:

memory: 2048MB

timeout: 540s

max_instances: 100

min_instances: 0

image-processing-handler:

memory: 4096MB

timeout: 300s

max_instances: 50

audit-log-processor:

memory: 1024MB

timeout: 60s

max_instances: 10

A.5 BigQuery Configuration

Dataset Structure

yaml

BigQuery_Datasets:

sarhi_operational:

location: US

default_table_expiration: null

default_partition_expiration: 2555 # 7 years

tables:

- patient_records
- clinical_notes
- billing_transactions
- audit_logs

access_controls:

- role: READER

service_account: sarhi-analytics@sarhi-prod-data.iam

- role: WRITER

service_account: sarhi-data-pipeline@sarhi-prod-data.iam

sarhi_analytics:

location: US

tables:

- patient_demographics_aggregate
- clinical_outcomes
- revenue_metrics
- operational_kpis

materialized_views:

- daily_patient_summary
- monthly_revenue_report
- provider_productivity

sarhi_ml:

location: US

tables:

- training_data
- model_predictions
- feature_store

A.6 Monitoring & Alerting Setup

Cloud Monitoring Configuration

yaml

Monitoring_Setup:

uptime_checks:

- name: api-gateway-health
monitored_resource: sarthi-api-gateway

http_check:

path: /health

port: 443

use_ssl: true

period: 60s

- name: database-connectivity
monitored_resource: cloudsql-database

tcp_check:

port: 5432

period: 60s

alert_policies:

- name: high-error-rate
condition:
metric: cloud_run_revision/request_count
filter: response_code_class="5xx"
threshold: 0.01 # 1% error rate
duration: 300s
- name: database-cpu-high
condition:
metric: cloudsql.googleapis.com/database/cpu/utilization
threshold: 0.8 # 80% CPU
duration: 600s
- name: storage-quota-exceeded
condition:
metric: storage.googleapis.com/storage/total_bytes
threshold: 1099511627776 # 1TB

notification_channels:

- type: email
labels:
email_address: ops-team@sarthi.healthcare
- type: sms
labels:
number: "+1-555-0100"
- type: pagerduty

labels:

service_key: \${PAGERDUTY_KEY}

A.7 Security & Compliance Infrastructure

Cloud KMS Setup

yaml

KMS_Configuration:

key_rings:

- name: sarthi-kr

location: us

keys:

- name: database-key

purpose: ENCRYPT_DECRYPT

rotation_period: 7776000s # 90 days

protection_level: HSM

- name: storage-key

purpose: ENCRYPT_DECRYPT

rotation_period: 7776000s

protection_level: HSM

- name: application-key

purpose: ENCRYPT_DECRYPT

rotation_period: 2592000s # 30 days

protection_level: SOFTWARE

VPC Service Controls

yaml

Service_Perimeter:

name: sarthi-healthcare-perimeter

resources:

- projects/12345 # *sarthi-prod-core*
- projects/12346 # *sarthi-prod-data*
- projects/12347 # *sarthi-prod-ai*

restricted_services:

- storage.googleapis.com
- bigquery.googleapis.com
- healthcare.googleapis.com
- aiplatform.googleapis.com

access_levels:

- **name:** corporate-access

conditions:

- **ip_subnetworks:**
 - 203.0.113.0/24 # *Corporate IP range*
- **device_policy:**
 - require_screen_lock:** true
 - require_corp_owned:** true

ingress_policies:

- **from:**
 - sources:**
 - **access_level:** corporate-access
- to:**
 - operations:**
 - **service_name:** storage.googleapis.com
 - method_selectors:**
 - **method:** google.storage.v1.Storage.GetObject

A.8 Cost Estimates & Sizing

Monthly Cost Projections (Production)

yaml

Cost_Estimates:

Compute:

Cloud_Run: \$2,500 # Based on 10M requests/month

Cloud_Functions: \$500

Storage:

Cloud_SQL: \$1,500 # HA instance + replicas

Cloud_Storage: \$800 # 5TB stored

BigQuery: \$1,000 # 10TB processed/month

Firestore: \$300 # 1M reads/writes per day

Networking:

Load_Balancing: \$200

Cloud_CDN: \$150

Cloud_NAT: \$100

AI_ML:

Document_AI: \$2,000 # 100k pages/month

Vertex_AI: \$1,500 # Custom model serving

Gemini_APIs: \$2,000 # Pro and Flash usage

Firebase:

Hosting: \$200

Authentication: \$500 # 10k MAU

Security:

Cloud_Armor: \$100

KMS: \$50

Total_Monthly: ~\$13,400

Environment_Multipliers:

Production: 1.0x

Staging: 0.5x # \$6,700

Development: 0.1x # \$1,340

A.9 Deployment Scripts

Initial Infrastructure Deployment

```
bash
```

```
#!/bin/bash
```

```
# deploy-infrastructure.sh
```

```
# Set project
```

```
export PROJECT_ID="sarathi-prod-core"
```

```
export REGION="us-central1"
```

```
# Create VPC and subnets
```

```
gcloud compute networks create sarathi-vpc \
```

```
--subnet-mode=custom \
```

```
--bgp-routing-mode=regional
```

```
gcloud compute networks subnets create sarathi-subnet-us-central1 \
```

```
--network=sarathi-vpc \
```

```
--region=${REGION} \
```

```
--range=10.0.0.0/20 \
```

```
--enable-private-ip-google-access \
```

```
--enable-flow-logs
```

```
# Create Cloud SQL instance
```

```
gcloud sql instances create sarathi-main-db \
```

```
--database-version=POSTGRES_15 \
```

```
--tier=db-custom-8-32768 \
```

```
--region=${REGION} \
```

```
--network=projects/${PROJECT_ID}/global/networks/sarathi-vpc \
```

```
--no-assign-ip \
```

```
--backup-start-time=02:00 \
```

```
--availability-type=REGIONAL
```

```
# Create Storage buckets
```

```
gsutil mb -c STANDARD -l US -b on gs://sarathi-patient-documents-prod
```

```
gsutil versioning set on gs://sarathi-patient-documents-prod
```

```
gsutil lifecycle set lifecycle.json gs://sarathi-patient-documents-prod
```

```
# Deploy Cloud Run services
```

```
gcloud run deploy sarathi-api-gateway \
```

```
--image=gcr.io/${PROJECT_ID}/sarathi-api-gateway:latest \
```

```
--platform=managed \
```

```
--region=${REGION} \
```

```
--min-instances=3 \
```

```
--max-instances=100 \
```

```
--memory=4Gi \
```

```
--cpu=2 \
```

```
--timeout=300 \  
--concurrency=250 \  
--service-account=sarathi-api-gateway@${PROJECT_ID}.iam.gserviceaccount.com
```

```
# Configure Firebase
```

```
firebase init hosting
```

```
firebase init firestore
```

```
firebase init storage
```

```
firebase deploy
```

A.10 Terraform Module Structure

Recommended Terraform Organization

```
yaml
```

terraform_modules:

root:

- main.tf
- variables.tf
- outputs.tf
- terraform.tfvars

modules:

networking:

- vpc.tf
- subnets.tf
- firewall.tf
- nat.tf

compute:

- cloud_run.tf
- cloud_functions.tf
- load_balancer.tf

data:

- cloud_sql.tf
- bigquery.tf
- storage.tf
- firestore.tf

security:

- iam.tf
- kms.tf
- service_perimeter.tf

monitoring:

- alerts.tf
- dashboards.tf
- uptime_checks.tf

firebase:

- hosting.tf
- auth.tf
- remote_config.tf

Appendix B: Data Migration & EHR Integration Patterns

B.1 EHR/EMR Integration Specifications

Epic Integration

yaml

Epic_Integration:

connection_method: FHIR R4 API

authentication: OAuth 2.0 with JWT

endpoints:

base_url: https://epic-hospital.org/fhir/api

auth_endpoint: /oauth2/token

resources_accessed:

- Patient
- Encounter
- Condition
- MedicationRequest
- Observation
- AllergyIntolerance
- DocumentReference

data_sync:

method: Webhook + Polling Hybrid

webhook_events:

- patient.created
- patient.updated
- encounter.closed
- result.final

polling_interval: 15 minutes

field_mappings:

epic_to_sarhi:

- epic: "Patient.identifier[?system='MRN'].value"
sarhi: "patient.medical_record_number"
- epic: "Patient.name[0].given + ' ' + Patient.name[0].family"
sarhi: "patient.full_name"
- epic: "Patient.birthDate"
sarhi: "patient.date_of_birth"
- epic: "Patient.telecom[?system='phone'].value"
sarhi: "patient.primary_phone"

error_handling:

retry_strategy: Exponential backoff with max 5 retries

failed_sync_queue: Dead letter queue in Pub/Sub

alert_threshold: 10 consecutive failures

yaml

Cerner_Integration:

connection_method: Cerner Millennium FHIR

authentication: SMART on FHIR

endpoints:

base_url: https://fhir.cerner.com/r4

auth_endpoint: /oauth2/authorize

token_endpoint: /oauth2/token

scopes_required:

- patient/*.read
- user/*.read
- launch
- openid
- profile

data_sync:

method: Subscription-based

subscription_criteria:

- Patient?_lastUpdated=gt{last_sync}
- Encounter?_lastUpdated=gt{last_sync}
- Observation?_lastUpdated=gt{last_sync}

special_considerations:

- Cerner-specific extensions for provider notes
- Custom coding systems for internal classifications
- Encounter-centric data model requiring joins

Athenahealth Integration

yaml

Athena_Integration:

connection_method: AthenaNet API v2

authentication: OAuth 2.0 Client Credentials

endpoints:

base_url: https://api.platform.athenahealth.com

api_patterns:

- /patients/{patientid}
- /appointments/{appointmentid}
- /encounters/{encounterid}
- /chart/{patientid}/problems
- /chart/{patientid}/medications

rate_limits:

requests_per_second: 10

daily_limit: 100000

data_transformation:

format: Custom JSON to FHIR R4

transformation_service: Cloud Functions

mapping_config: gs://sarathi-configs/athena-fhir-mappings.json

B.2 HL7 FHIR Transformation Pipelines

FHIR Data Pipeline Architecture

yaml

Pipeline_Configuration:

ingestion_layer:

cloud_healthcare_api:

fhir_store: projects/sarthi-prod/locations/us-central1/datasets/ehr-integration/fhirStores/main

validation:

- FHIR R4 structure validation
- Required field presence
- Referential integrity
- Terminology binding validation

transformation_layer:

dataflow_job:

template: FHIR_to_BigQuery_Streaming

transformations:

- Flatten nested FHIR resources
- Normalize code systems
- Derive calculated fields
- Apply business rules

code_system_mapping:

ICD10:

source: <http://hl7.org/fhir/sid/icd-10>

target: internal_diagnosis_codes

LOINC:

source: <http://loinc.org>

target: internal_lab_codes

SNOMED:

source: <http://snomed.info/sct>

target: internal_clinical_terms

output_layer:

destinations:

- **BigQuery:** Analytical data warehouse
- **Cloud SQL:** Operational database
- **Pub/Sub:** Real-time event streaming

B.3 Legacy Data Migration Procedures

Migration Strategy

yaml

Migration_Phases:

Phase_1_Assessment:

duration: 2 weeks

activities:

- Data quality assessment
- Volume analysis
- Dependency mapping
- Risk identification

deliverables:

- Data inventory spreadsheet
- Quality score report
- Migration complexity matrix

Phase_2_Preparation:

duration: 3 weeks

activities:

- Schema mapping
- Transformation rules development
- Validation rules creation
- Rollback procedures

Phase_3_Test_Migration:

duration: 2 weeks

activities:

- 10% sample migration
- Validation execution
- Performance testing
- Issue resolution

Phase_4_Production_Migration:

duration: 1 week

activities:

- Full data migration
- Parallel run validation
- Cutover execution
- Post-migration verification

Data Validation & Reconciliation

yaml

Validation_Framework:

pre_migration_checks:

- Row count verification
- Checksum validation
- Foreign key integrity
- Date range validation

transformation_validation:

- Field mapping accuracy
- Data type conversion
- Null handling
- Default value application

post_migration_validation:

- Row count matching
- Sum/aggregate matching
- Random sample verification
- Business rule validation

reconciliation_reports:

- Source vs target row counts
- Failed record details
- Transformation exceptions
- Performance metrics

Appendix C: Disaster Recovery Runbooks

C.1 Failover Procedures

Database Failover Runbook

yaml

Primary_Database_Failure:

detection:

- Cloud Monitoring alert triggered
- Health check failures > 3 consecutive
- Replication lag > 30 seconds

automatic_actions:

- Notification sent to on-call
- Replica promotion initiated
- Application connection redirect

manual_steps:

1_verify_failure:

command: |

```
gcloud sql instances describe sarthi-main-db \
  --format="get(state,replicaConfiguration.failoverTarget)"
```

expected: "FAILED" or unreachable

2_promote_replica:

command: |

```
gcloud sql instances promote-replica sarthi-main-db-replica-1
```

wait_time: 2-5 minutes

3_update_dns:

command: |

```
gcloud dns record-sets update db.sarthi.internal \
  --type=A \
  --rrdatas=10.1.0.5 \
  --zone=sarthi-internal-zone
```

4_verify_applications:

checks:

- API Gateway health endpoint
- Database connection pool status
- Query execution test

5_update_status_page:

url: <https://status.sarthi.healthcare>

message: "Database failover completed successfully"

Regional Failover Runbook

Region_Failure_Response:

trigger_conditions:

- Multiple service failures in region
- Network partition detected
- Natural disaster warning

failover_sequence:

1_initiate_traffic_shift:

steps:

- Update global load balancer backend priorities
- Shift traffic weights to secondary region
- Monitor error rates during transition

2_database_failover:

- Promote regional replica to primary
- Update connection strings
- Verify replication to new primary

3_cache_warming:

- Pre-populate Redis cache in secondary region
- Sync session data
- Update cache references

4_notification:

- Update status page
- Notify customers via email
- Internal team alerts

C.2 Recovery Objectives

Service-Level Recovery Targets

yaml

Recovery_Objectives:

tier_1_critical:

services:

- Patient portal login
- Clinical documentation
- Medication management

RTO: 15 minutes

RPO: 5 minutes

tier_2_essential:

services:

- Appointment scheduling
- Billing processing
- Lab result viewing

RTO: 1 hour

RPO: 15 minutes

tier_3_standard:

services:

- Analytics dashboards
- Report generation
- Inventory management

RTO: 4 hours

RPO: 1 hour

C.3 DR Testing Procedures

Quarterly DR Test Plan

yaml

DR_Test_Schedule:

Q1_Test:

type: Tabletop exercise

scope: Full regional failure scenario

participants: All engineering teams

duration: 4 hours

Q2_Test:

type: Database failover

scope: Primary database failure

environment: Staging

duration: 2 hours

Q3_Test:

type: Application failover

scope: Cloud Run service failures

environment: Production (controlled)

duration: 1 hour

Q4_Test:

type: Full regional failover

scope: Complete regional failure

environment: Production (off-hours)

duration: 6 hours

success_criteria:

- Meet RTO/RPO targets
- No data loss
- Successful rollback
- Documentation updates completed

Appendix D: CI/CD Pipeline Configuration

D.1 Cloud Build Pipeline

Production Deployment Pipeline

yaml

```
# cloudbuild.yaml
```

steps:

```
# Run security scanning
```

```
- name: 'gcr.io/cloud-builders/gcloud'
```

```
  entrypoint: 'bash'
```

```
  args:
```

```
    - '-c'
```

```
    - |
```

```
      # Dependency vulnerability scanning
```

```
      npm audit --audit-level=high
```

```
# SAST scanning
```

```
docker run --rm -v $(pwd):/src \
```

```
  returntocorp/semgrep:latest \
```

```
  --config=auto --severity=ERROR /src
```

```
# Container scanning
```

```
gcloud container images scan ${_IMAGE_NAME}:${SHORT_SHA}
```

```
# Run unit tests
```

```
- name: 'node:20'
```

```
  entrypoint: 'npm'
```

```
  args: ['test']
```

```
  env:
```

```
    - 'NODE_ENV=test'
```

```
# Run integration tests
```

```
- name: 'node:20'
```

```
  entrypoint: 'npm'
```

```
  args: ['run', 'test:integration']
```

```
# Build application
```

```
- name: 'gcr.io/kaniko-project/executor:latest'
```

```
  args:
```

```
    - '--dockerfile=Dockerfile'
```

```
    - '--cache=true'
```

```
    - '--cache-ttl=24h'
```

```
    - '--destination=gcr.io/${PROJECT_ID}/${_SERVICE_NAME}:${SHORT_SHA}'
```

```
    - '--destination=gcr.io/${PROJECT_ID}/${_SERVICE_NAME}:latest'
```

```
# Deploy to staging
```

```
- name: 'gcr.io/cloud-builders/gcloud'
```

```
  args:
```

```
- 'run'
- 'deploy'
- '${_SERVICE_NAME}'
- '--image=gcr.io/${PROJECT_ID}/${_SERVICE_NAME}:${SHORT_SHA}'
- '--region=us-central1'
- '--platform=managed'
- '--tag=staging'
- '--no-traffic'
```

Run smoke tests

```
- name: 'node:20'
  entrypoint: 'npm'
  args: ['run', 'test:smoke']
  env:
    - 'TARGET_URL=https://staging---${_SERVICE_NAME}-abc123.a.run.app'
```

Manual approval gate (for production)

```
- name: 'gcr.io/cloud-builders/gcloud'
  entrypoint: 'bash'
  args:
    - '-c'
    - |
      if [ "${BRANCH_NAME}" == "main" ]; then
        echo "Waiting for manual approval..."
        gcloud builds approve ${BUILD_ID}
      fi
  waitFor: ['-']
```

Production deployment

```
- name: 'gcr.io/cloud-builders/gcloud'
  args:
    - 'run'
    - 'services'
    - 'update-traffic'
    - '${_SERVICE_NAME}'
    - '--region=us-central1'
    - '--to-tags=staging=10' # Canary deployment
```

Monitor canary metrics

```
- name: 'gcr.io/${PROJECT_ID}/canary-validator:latest'
  args:
    - '--service=${_SERVICE_NAME}'
    - '--duration=600'
    - '--error-rate-threshold=0.01'
```

```
- '--latency-threshold=500'
```

```
# Full production rollout
```

```
- name: 'gcr.io/cloud-builders/gcloud'
```

```
args:
```

```
- 'run'
```

```
- 'services'
```

```
- 'update-traffic'
```

```
- '${_SERVICE_NAME}'
```

```
- '--region=us-central1'
```

```
- '--to-latest=100'
```

```
substitutions:
```

```
_SERVICE_NAME: sarthi-api-gateway
```

```
_IMAGE_NAME: gcr.io/${PROJECT_ID}/sarthi-api-gateway
```

```
options:
```

```
logging: CLOUD_LOGGING_ONLY
```

```
machineType: 'E2_HIGHCPU_8'
```

```
timeout: '2400s'
```

D.2 GitHub Actions Integration

GitHub Actions Workflow

```
yaml
```

.github/workflows/deploy.yml

name: Deploy to GCP

on:

push:

branches: [main, staging, develop]

pull_request:

branches: [main]

env:

PROJECT_ID: sarthi-prod-core

GAR_LOCATION: us-central1

jobs:

security-scan:

runs-on: ubuntu-latest

steps:

- uses: actions/checkout@v3
- name: Run Trivy vulnerability scanner
uses: aquasecurity/trivy-action@master
with:
scan-type: 'fs'
scan-ref: '.'
severity: 'CRITICAL,HIGH'
- name: SAST with CodeQL
uses: github/codeql-action/analyze@v2
- name: Secret scanning
uses: trufflesecurity/trufflehog@main
with:
path: ./

test:

runs-on: ubuntu-latest

needs: security-scan

services:

postgres:

image: postgres:15

env:

POSTGRES_PASSWORD: testpass

options: >-

- health-cmd pg_isready
- health-interval 10s
- health-timeout 5s
- health-retries 5

steps:

- uses: actions/checkout@v3
- uses: actions/setup-node@v3

with:

node-version: '20'
cache: 'npm'

- run: npm ci
- run: npm test -- --coverage
- run: npm run test:integration

- name: Upload coverage
uses: codecov/codecov-action@v3
with:
threshold: 80%

deploy:

needs: [security-scan, test]
runs-on: ubuntu-latest
if: github.ref == 'refs/heads/main'

steps:

- uses: actions/checkout@v3
- uses: google-github-actions/auth@v1
with:
credentials_json: \${{ secrets.GCP_SA_KEY }}
- name: Set up Cloud SDK
uses: google-github-actions/setup-gcloud@v1
- name: Trigger Cloud Build
run: |
gcloud builds submit \
--config=cloudbuild.yaml \
--substitutions=BRANCH_NAME=\${{ github.ref_name }}

D.3 Environment Promotion

Promotion Gates Configuration

yaml

Environment_Promotion:

dev_to_staging:

automatic: true

conditions:

- All tests passing
- Code coverage > 80%
- No critical vulnerabilities
- Performance benchmarks met

staging_to_production:

automatic: false

approval_required:

- Engineering lead
- Product owner
- Security team (for major changes)

pre_production_checklist:

- Load test completed
- Security scan passed
- Rollback plan documented
- Monitoring alerts configured
- Status page updated

deployment_window:

days: [Tuesday, Thursday]

time: "02:00-06:00 UTC"

blackout_dates:

- Major holidays
- End of month (billing)
- Scheduled maintenance windows

Appendix E: Secret Management Architecture

E.1 Secret Rotation Schedule

Automated Secret Rotation

yaml

Secret_Rotation_Policy:

database_passwords:

rotation_period: 30 days

notification: 7 days before expiry

process:

1_create_new_password:

service: Cloud KMS

strength: 32 characters, mixed case, symbols

2_update_database:

command: |

```
gcloud sql users set-password ${USER} \  
--instance=${INSTANCE} \  
--password=${NEW_PASSWORD}
```

3_update_secret_manager:

command: |

```
echo -n ${NEW_PASSWORD} | gcloud secrets versions add ${SECRET_NAME} \  
--data-file=-
```

4_trigger_deployment:

action: Rolling restart of dependent services

5_verify_connectivity:

health_checks: All services report healthy

6_cleanup_old_version:

delay: 24 hours

action: Disable old secret version

api_keys:

rotation_period: 90 days

third_party_apis:

- stripe: Manual rotation via dashboard
- twilio: API-based rotation
- sendgrid: Automated via API

service_account_keys:

rotation_period: 90 days

policy: Use workload identity where possible

ssl_certificates:

rotation_period: 60 days before expiry

automation: Cert-manager with Let's Encrypt

E.2 Certificate Management

SSL/TLS Certificate Lifecycle

yaml

Certificate_Management:

managed_certificates:

provider: Google-managed SSL

domains:

- "*.sarthi.healthcare"
- "api.sarthi.healthcare"
- "app.sarthi.healthcare"

auto_renewal: true

custom_certificates:

storage: Secret Manager

renewal_process:

- Monitor expiry via Cloud Monitoring
- Generate CSR 30 days before expiry
- Submit to Certificate Authority
- Validate domain ownership
- Deploy new certificate
- Update load balancer
- Verify SSL labs score

internal_certificates:

ca: Private CA in Cloud Certificate Authority Service

validity: 365 days

subjects:

- Internal microservices
- Database connections
- Service mesh mTLS

E.3 Encryption Key Hierarchy

Key Management Structure

yaml

Key_Hierarchy:

root_keys:

level: Hardware Security Module (HSM)
location: Cloud KMS
purpose: Master key encryption
access: Break-glass procedure only

master_keys:

level: Cloud KMS
rotation: 90 days

types:

- Database encryption key
- Storage encryption key
- Application secrets key
- Backup encryption key

data_encryption_keys:

level: Application
rotation: 30 days
derivation: HKDF from master keys

usage:

- PHI field encryption
- Session token encryption
- Temporary file encryption

key_derivation:

algorithm: HKDF-SHA256

context_binding:

- Tenant ID
- Resource type
- Timestamp
- Version

Appendix F: API Gateway Configuration

F.1 Rate Limiting Rules

Rate Limit Configuration

yaml

Rate_Limiting:

global_limits:

per_second: 10000

per_minute: 300000

burst_size: 1000

per_tenant_limits:

tiers:

starter:

per_second: 100

per_minute: 3000

concurrent_requests: 50

professional:

per_second: 500

per_minute: 15000

concurrent_requests: 200

enterprise:

per_second: 2000

per_minute: 60000

concurrent_requests: 1000

per_endpoint_limits:

"/api/v1/auth/login":

per_minute: 10

per_ip: true

"/api/v1/documents/upload":

per_minute: 60

max_file_size: 50MB

"/api/v1/ai/clinical-note":

per_minute: 30

queue_overflow: 503 Service Unavailable

rate_limit_headers:

- X-RateLimit-Limit

- X-RateLimit-Remaining

- X-RateLimit-Reset

- Retry-After (on 429 responses)

F.2 API Versioning Strategy

Version Management

yaml

API_Versioning:

strategy: URI versioning with header override

current_versions:

stable: v1

beta: v2-beta

deprecated: v0 (sunset date: 2025-12-31)

version_routing:

default: v1

header_override:

header_name: X-API-Version

accepted_values: [v1, v2-beta]

deprecation_policy:

notice_period: 6 months

deprecation_timeline:

T-180_days: Deprecation announced

T-90_days: Console warnings added

T-30_days: Deprecation headers in responses

T-7_days: Daily reminder emails

T-0: Version sunset

backward_compatibility:

minor_versions: Must be backward compatible

major_versions: Breaking changes allowed

compatibility_tests:

- Contract testing
- Response format validation
- Field presence checking

F.3 Request/Response Transformation

Data Transformation Rules

yaml

Transformations:

request_transformations:

tenant_injection:

source: JWT token claim

target: X-Tenant-ID header

date_normalization:

input_formats:

- ISO 8601
- MM/DD/YYYY
- Unix timestamp

output_format: ISO 8601

pii_masking:

fields:

- ssn: XXX-XX-**{last4}**
- credit_card: XXXX-XXXX-XXXX-**{last4}**
- dob: {year}-XX-XX (for logging)

response_transformations:

field_filtering:

based_on: User role and permissions

pagination:

default_page_size: 50

max_page_size: 200

wrapper_format:

data: [...results...]

meta:

page: 1

per_page: 50

total: 1000

total_pages: 20

links:

first: /api/v1/resource?page=1

last: /api/v1/resource?page=20

next: /api/v1/resource?page=2

G.1 Structured Logging Format

Standard Log Schema

yaml

Log_Schema:

required_fields:

timestamp: ISO 8601 with timezone

level: ERROR|WARN|INFO|DEBUG

service: Service identifier

trace_id: Correlation ID for distributed tracing

span_id: Current operation ID

tenant_id: Multi-tenant identifier

contextual_fields:

user_id: Authenticated user

session_id: User session

request_id: HTTP request ID

method: HTTP method

path: Request path

status_code: HTTP response code

duration_ms: Request duration

error_fields:

error_type: Exception class

error_message: Error description

stack_trace: Full stack (dev only)

error_code: Application error code

json_format_example: |

```
{
  "timestamp": "2025-01-15T10:30:45.123Z",
  "level": "ERROR",
  "service": "sarathi-api-gateway",
  "trace_id": "abc123def456",
  "span_id": "789ghi",
  "tenant_id": "tenant_001",
  "user_id": "user_12345",
  "method": "POST",
  "path": "/api/v1/clinical-notes",
  "status_code": 500,
  "duration_ms": 1234,
  "error_type": "DatabaseConnectionError",
  "error_message": "Unable to connect to primary database",
  "error_code": "DB_CONN_001"
}
```

G.2 PHI Detection & Redaction

PHI Redaction Rules

yaml

PHI_Detection:

detection_methods:

regex_patterns:

ssn: '\b\d{3}-?\d{2}-?\d{4}\b'

mrn: '\b[A-Z]{2}\d{6,8}\b'

phone: '\b\d{3}[-.]?\d{3}[-.]?\d{4}\b'

email: '\b[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Z|a-z]{2,}\b'

dob: '\b(0[1-9]|1[0-2])/.[0-9]|(0[1-9]|12)\d3[01])/.[0-9]|(19|20)\d2\b'

ml_detection:

service: Cloud DLP API

info_types:

- PERSON_NAME
- DATE_OF_BIRTH
- US_SOCIAL_SECURITY_NUMBER
- PHONE_NUMBER
- EMAIL_ADDRESS
- US_HEALTHCARE_NPI

custom_dictionaries:

- Medical conditions
- Medication names
- Provider names

redaction_strategy:

logs:

method: Replace with [REDACTED-{TYPE}]

analytics:

method: Tokenization with secure lookup

debugging:

method: Partial masking (show first/last characters)

exemptions:

- Audit logs (stored separately with encryption)
- Error reports (PHI removed before transmission)

G.3 Audit Log Requirements

HIPAA Audit Logging

yaml

Audit_Requirements:

required_events:

authentication:

- Login attempts (success/failure)
- Logout events
- Password changes
- MFA challenges

authorization:

- Permission grants/revokes
- Role assignments
- Access denials

data_access:

- Patient record views
- Report generation
- Data exports
- Print operations

data_modification:

- Create operations
- Update operations
- Delete operations
- Bulk operations

system_events:

- Configuration changes
- Security policy updates
- System starts/stops
- Backup/restore operations

audit_log_format:

minimum_fields:

- Event timestamp
- User identification
- Event type
- Resource accessed
- Source IP address
- Success/failure indicator
- Reason for failure

retention:

audit_logs: 7 years

access_logs: 2 years

error_logs: 1 year

debug_logs: 30 days

protection:

- Immutable storage
- Encryption at rest
- Tamper detection
- Separate access controls

Appendix H: Performance Testing Scenarios

H.1 Load Test Specifications

Load Test Scenarios

yaml

Load_Test_Suite:

baseline_test:

duration: 30 minutes
virtual_users: 100
ramp_up: 5 minutes

scenarios:

- Login and dashboard view: 40%
- Patient record access: 30%
- Document upload: 20%
- Report generation: 10%

stress_test:

duration: 1 hour
virtual_users: Start 100, increase by 100 every 10 min
target: Find breaking point

metrics_monitored:

- Response time percentiles (p50, p95, p99)
- Error rate
- Throughput
- Resource utilization

spike_test:

duration: 30 minutes
pattern: 100 → 1000 → 100 users
spike_duration: 5 minutes

validation:

- System recovers to baseline
- No data corruption
- Queue drainage successful

endurance_test:

duration: 24 hours
virtual_users: 500

checks:

- Memory leaks
- Connection pool exhaustion
- Disk space growth
- Log rotation

H.2 Performance Baselines

Expected Performance Metrics

yaml

Performance_SLIs:

api_endpoints:

GET_/api/v1/patients/{id}:

p50_latency: 50ms
p95_latency: 200ms
p99_latency: 500ms
error_rate: < 0.1%

POST_/api/v1/documents/upload:

p50_latency: 2s
p95_latency: 5s
p99_latency: 10s
throughput: 100 req/min

POST_/api/v1/ai/clinical-note:

p50_latency: 3s
p95_latency: 7s
p99_latency: 15s
throughput: 50 req/min

database_operations:

simple_select:

latency: < 10ms

complex_join:

latency: < 100ms

bulk_insert:

throughput: 10000 rows/sec

ai_operations:

document_ocr:

pages_per_minute: 60
accuracy: > 95%

clinical_note_generation:

tokens_per_second: 100
latency: < 5s

H.3 Capacity Planning Models

Growth Projections

yaml

Capacity_Model:

current_metrics:

daily_active_users: 1000

peak_concurrent_users: 200

daily_transactions: 50000

storage_growth_rate: 100GB/month

growth_assumptions:

user_growth: 20% per quarter

transaction_growth: 25% per quarter

storage_growth: 30% per quarter

scaling_triggers:

cpu_utilization: > 70% for 10 minutes

memory_utilization: > 80% for 10 minutes

request_latency: p95 > 1 second

error_rate: > 1% for 5 minutes

capacity_thresholds:

6_month_projection:

users: 2000

infrastructure:

- Add 2 Cloud Run instances
- Upgrade database tier
- Increase cache size

12_month_projection:

users: 4000

infrastructure:

- Multi-region deployment
- Read replica databases
- CDN implementation

Appendix I: Network Connectivity for Healthcare Organizations

I.1 VPN Configuration

Site-to-Site VPN Setup

yaml

VPN_Configuration:

cloud_vpn_gateway:

name: sarthi-vpn-gateway

region: us-central1

interfaces:

- 0: 35.242.117.95

- 1: 35.220.73.140 # *Redundancy*

vpn_tunnels:

hospital_network_primary:

peer_gateway: 198.51.100.1

ike_version: 2

shared_secret: \${SECRET_MANAGER_REF}

phase1_params:

encryption: AES-256

integrity: SHA-256

dh_group: 14

lifetime: 36000

phase2_params:

encryption: AES-256

integrity: SHA-256

pfs_group: 14

lifetime: 10800

hospital_network_backup:

peer_gateway: 198.51.100.2

configuration: Same as primary

routing:

advertised_routes:

- 10.0.0.0/16 # *Sarthi VPC*

learned_routes:

- 192.168.0.0/16 # *Hospital network*

monitoring:

- Tunnel status

- Packet loss

- Latency

- Bandwidth utilization

I.2 Direct Connect Setup

Dedicated Interconnect

yaml

Interconnect_Configuration:

connection:

type: Dedicated Interconnect

capacity: 10 Gbps

location: Chicago edge location

vlan_attachments:

primary:

vlan_id: 1000

bandwidth: 5 Gbps

region: us-central1

secondary:

vlan_id: 1001

bandwidth: 5 Gbps

region: us-east1

bgp_configuration:

cloud_router:

asn: 64512

advertised_routes:

- 10.0.0.0/8

keepalive_interval: 20

customer_router:

peer_asn: 64513

peer_ip: 169.254.1.1

security:

- MACsec encryption
- BGP MD5 authentication
- IP allowlisting

I.3 Network Segmentation

Security Zones

yaml

Network_Zones:

dmz_zone:

subnet: 10.0.1.0/24

allowed_services:

- Load balancers
- WAF
- DDoS protection

firewall_rules:

- Allow 443 from 0.0.0.0/0
- Allow 80 from 0.0.0.0/0 (redirect to 443)
- Deny all other inbound

application_zone:

subnet: 10.0.2.0/24

allowed_services:

- Cloud Run services
- Cloud Functions
- API Gateway

firewall_rules:

- Allow from DMZ zone
- Allow from management zone
- Deny from internet

data_zone:

subnet: 10.0.3.0/24

allowed_services:

- Cloud SQL
- Cloud Storage
- BigQuery

firewall_rules:

- Allow from application zone only
- Deny all other

management_zone:

subnet: 10.0.4.0/24

allowed_services:

- Bastion hosts
- Monitoring tools
- CI/CD systems

access_control:

- VPN required
- MFA enforced
- Session recording enabled

Appendix J: Incident Response Procedures

J.1 Incident Classification

Severity Levels

```
yaml
```

Incident_Severity:

P1_Critical:

definition: Complete service outage or data breach

examples:

- Database failure affecting all users
- Security breach detected
- PHI data exposure
- Complete region failure

response_time: 15 minutes

escalation: Immediate

communication: All stakeholders + customers

P2_Major:

definition: Significant degradation affecting many users

examples:

- API response times > 5 seconds
- Authentication service issues
- Partial feature outage

response_time: 30 minutes

escalation: After 1 hour

communication: Internal + status page

P3_Minor:

definition: Limited impact affecting small user subset

examples:

- Report generation delays
- Non-critical feature issues
- Isolated performance degradation

response_time: 2 hours

escalation: After 4 hours

communication: Internal only

P4_Low:

definition: No immediate user impact

examples:

- Non-production issues

- Documentation problems
- Minor UI glitches

response_time: Next business day

escalation: Not required

communication: Ticket system

J.2 Escalation Matrix

Escalation Paths

yaml

Escalation_Chain:

on_call_rotation:

primary:

schedule: Weekly rotation

regions: Follow the sun model

backup:

activation: No response in 15 minutes

escalation_levels:

L1_on_call_engineer:

response_time: 15 minutes

responsibilities:

- Initial assessment
- Basic troubleshooting
- Incident commander for P3/P4

L2_senior_engineer:

response_time: 30 minutes

trigger: P1/P2 or L1 escalation

responsibilities:

- Advanced troubleshooting
- Coordinate fixes
- Customer communication

L3_engineering_manager:

response_time: 45 minutes

trigger: P1 or extended P2

responsibilities:

- Resource allocation
- Executive communication
- Major decision authority

L4_cto:

response_time: 1 hour

trigger: Data breach or extended P1

responsibilities:

- Crisis management
- External communication
- Legal coordination

J.3 Communication Templates

Incident Communications

yaml

Communication_Templates:

initial_notification:

subject: "[P{severity}] Incident Detected: {brief_description}"

body: |

Incident Start Time: {timestamp}

Severity: P{severity}

Affected Service: {service}

Impact: {user_impact}

Current Status: Investigating

Incident Commander: {ic_name}

Conference Bridge: {bridge_info}

Next Update: In 30 minutes or when status changes

customer_notification:

subject: "Service Disruption Notice"

body: |

Dear Customer,

We are currently experiencing {general_description}.

Impact: {customer_impact}

Affected Services: {services_list}

Our team is actively working to resolve this issue.

For updates: <https://status.sarthi.healthcare>

We apologize for any inconvenience.

resolution_notice:

subject: "[RESOLVED] P{severity} Incident: {description}"

body: |

Incident Duration: {start_time} - {end_time}

Total Time: {duration}

Root Cause: {brief_root_cause}

Resolution: {resolution_action}

Follow-up Actions:

- Post-mortem scheduled for {date}
- Preventive measures to be implemented

Full RCA will be available within 48 hours.

J.4 Post-Incident Process

Root Cause Analysis Template

yaml

Post_Mortem_Template:

incident_summary:

- Incident ID
- Date and duration
- Severity level
- Services affected
- Customer impact metrics

timeline:

- Detection time and method
- Key events and actions
- Resolution time
- Service restoration

root_cause_analysis:

- Primary cause
- Contributing factors
- Why it wasn't prevented
- Why it wasn't detected sooner

impact_assessment:

- Users affected
- Data impact
- Financial impact
- Reputation impact

what_went_well:

- Effective responses
- Good decisions
- Team coordination

what_went_wrong:

- Failed processes
- Missing alerts
- Communication gaps

action_items:

- priority: P0
- owner: Assigned individual
- deadline: Specific date
- description: Detailed action

lessons_learned:

- Process improvements
- Technical improvements
- Documentation updates

Appendix K: Compliance Validation Framework

K.1 HIPAA Technical Safeguards Checklist

Technical Controls Verification

yaml

HIPAA_Technical_Checklist:

access_control:

- ✓ Unique user identification
- ✓ Automatic logoff after 15 minutes
- ✓ Encryption and decryption
- ✓ Access authorization procedures

validation_method:

- Penetration testing
- Access review audit
- Configuration scanning

audit_controls:

- ✓ Hardware, software, procedural mechanisms
- ✓ Log examination capability
- ✓ Audit log protection
- ✓ 7-year retention

validation_method:

- Log integrity testing
- Retention policy audit
- Access control review

integrity_controls:

- ✓ PHI alteration detection
- ✓ PHI destruction prevention
- ✓ Electronic signature capability
- ✓ Data validation rules

validation_method:

- Data integrity testing
- Change tracking audit
- Backup verification

transmission_security:

- ✓ Network security controls
- ✓ Encryption in transit (TLS 1.2+)
- ✓ Integrity controls
- ✓ VPN for remote access

validation_method:

- SSL/TLS configuration scan

- Network traffic analysis
- Vulnerability assessment

K.2 Security Assessment Schedule

Assessment Calendar

yaml

Security_Assessments:

quarterly_assessments:

Q1:

- Vulnerability scanning (internal)
- Firewall rule review
- Access control audit
- Patch compliance check

Q2:

- Penetration testing (external)
- Social engineering test
- Disaster recovery drill
- Encryption audit

Q3:

- Vulnerability scanning (internal)
- Code security review
- Third-party risk assessment
- Compliance gap analysis

Q4:

- Full security audit
- Penetration testing (internal)
- Incident response drill
- Annual risk assessment

monthly_assessments:

- Security patch review
- Access review (privileged accounts)
- Security metrics review
- Threat intelligence briefing

continuous_monitoring:

- Real-time threat detection
- File integrity monitoring
- Network anomaly detection
- User behavior analytics

K.3 Compliance Reporting

Automated Compliance Dashboard

Compliance_Metrics:

dashboard_components:

security_posture:

metrics:

- Vulnerability count by severity
- Patch compliance percentage
- Security training completion
- Incident response times

access_management:

metrics:

- Orphaned accounts
- Privilege escalation events
- Failed login attempts
- MFA adoption rate

data_protection:

metrics:

- Encryption coverage
- Data classification status
- DLP policy violations
- Backup success rate

audit_compliance:

metrics:

- Audit log completeness
- Review completion rate
- Finding remediation time
- Control effectiveness

reporting_schedule:

daily:

- Security event summary
- Critical vulnerability alerts

weekly:

- Compliance scorecard
- Risk register updates

monthly:

- Executive dashboard
- Trend analysis report

quarterly:

- Board compliance report
- Regulatory filing preparation

K.4 Vulnerability Management

Vulnerability Response SLA

yaml

Vulnerability_Management:

severity_classification:

critical:

cvss_score: 9.0 - 10.0

remediation_sla: 24 hours

high:

cvss_score: 7.0 - 8.9

remediation_sla: 7 days

medium:

cvss_score: 4.0 - 6.9

remediation_sla: 30 days

low:

cvss_score: 0.1 - 3.9

remediation_sla: 90 days

scanning_schedule:

infrastructure:

frequency: Weekly

scope: All production systems

application:

frequency: Per deployment + monthly

scope: All web applications

container_images:

frequency: On build + daily

scope: All container registries

remediation_tracking:

- Discovery date
- Severity assessment
- Owner assignment
- Remediation deadline
- Validation testing
- Closure verification

Appendix L: Multi-Tenancy Implementation

L.1 Tenant Isolation Strategies

Database Isolation Models

yaml

Tenant_Isolation_Patterns:

row_level_security:

implementation: PostgreSQL RLS Policies

schema_design:

- Every table includes tenant_id column
- **Composite primary keys:** (tenant_id, entity_id)
- Foreign keys include tenant_id

rls_policies:

enable_rls: |

```
ALTER TABLE patients ENABLE ROW LEVEL SECURITY;
```

create_policy: |

```
CREATE POLICY tenant_isolation ON patients  
FOR ALL
```

```
USING (tenant_id = current_setting('app.current_tenant')::uuid);
```

connection_setup: |

```
SET app.current_tenant = 'tenant_uuid';
```

```
SET ROLE app_user;
```

advantages:

- Shared infrastructure cost efficiency
- Easy cross-tenant analytics
- Simple backup strategy

considerations:

- Careful index design needed
- Query performance monitoring
- Noisy neighbor potential

schema_separation:

implementation: Schema per tenant

naming_convention:

pattern: tenant_{tenant_id}

example: tenant_abc123

provisioning: |

```
CREATE SCHEMA tenant_${tenant_id};
```

```
GRANT USAGE ON SCHEMA tenant_${tenant_id} TO app_user;
```

-- Clone base schema

```
SELECT clone_schema('base_template', 'tenant_${tenant_id}');
```

connection_routing:

method: Dynamic schema switching

implementation: |

```
SET search_path TO tenant_${tenant_id}, public;
```

advantages:

- Stronger isolation
- Per-tenant customization
- Independent tenant backup/restore

hybrid_approach:

strategy: Tiered isolation based on plan

tiers:

starter:

model: Row-level security

resource_limits: Shared pool

professional:

model: Schema separation

resource_limits: Guaranteed minimums

enterprise:

model: Dedicated database

resource_limits: Dedicated resources

L.2 Tenant Onboarding Automation

Automated Provisioning Pipeline

yaml

Tenant_Onboarding:

workflow_steps:

1_initial_setup:

trigger: Signup completion

actions:

- Generate tenant UUID
- Create tenant record
- Set up billing account
- Send welcome email

2_infrastructure_provisioning:

parallel_execution: true

database_setup:

- Create schema/RLS policies
- Seed default data
- Configure connection pool

storage_setup:

- Create tenant folder structure
- Set IAM policies
- Configure quotas

identity_setup:

- Create identity provider tenant
- Configure SSO if requested
- Set up default roles

3_data_seeding:

default_data:

- System roles and permissions
- Workflow templates
- Report templates
- Default settings

sample_data:

enabled: Optional for trials

includes:

- Demo patients
- Sample appointments
- Example documents

4_configuration:

tenant_settings:

- Timezone configuration
- Business hours
- Holiday calendar
- Compliance settings

feature_flags:

- Enable trial features
- Set plan limitations
- Configure integrations

5_validation:

health_checks:

- Database connectivity
- API accessibility
- Storage permissions
- Authentication flow

notification:

- Send activation email
- Provide onboarding resources
- Schedule onboarding call

automation_tools:

orchestration: Cloud Workflows

workflow_definition: |

main:

params: [tenant_data]

steps:

- createTenant:

call: http.post

args:

url: \${api_base}/internal/tenants

body: \${tenant_data}

auth:

type: OIDC

- provisionDatabase:

call: provision_database

args:

tenant_id: \${createTenant.body.tenant_id}

- setupStorage:

call: setup_storage

```
  args:
    tenant_id: ${createTenant.body.tenant_id}
- sendNotification:
  call: send_email
  args:
    template: welcome
    recipient: ${tenant_data.admin_email}
```

L.3 Tenant Offboarding Procedures

Data Retention & Deletion

```
yaml
```

Tenant_Offboarding:

retention_policies:

healthcare_data:

period: 7 years

storage: Cold storage archive

format: Encrypted backup

operational_data:

period: 90 days

storage: Standard storage

format: JSON export

audit_logs:

period: 7 years

storage: BigQuery archive

format: Structured logs

offboarding_workflow:

1_initiation:

triggers:

- Cancellation request
- Non-payment (after grace period)
- Contract expiration

2_data_export:

formats_available:

- FHIR bundle
- CSV packages
- SQL dump
- Document archive

delivery_method:

- Secure download link
- SFTP transfer
- Cloud storage transfer

3_service_suspension:

grace_period: 30 days

actions:

- Disable user logins
- Stop background jobs
- Pause API access

- Display suspension notice

4_data_archival:

process:

- Create final backup
- Encrypt with customer key
- Move to cold storage
- Update data catalog

5_resource_cleanup:

immediate:

- Remove from load balancer
- Delete cache entries
- Release reserved capacity

after_retention:

- Delete archived data
- Remove encryption keys
- Purge audit logs
- Certificate of destruction

L.4 Cross-Tenant Analytics

Privacy-Preserving Analytics

yaml

Cross_Tenant_Analytics:

aggregation_rules:

minimum_cohort_size: 10

anonymization:

- Remove direct identifiers
- Generalize quasi-identifiers
- Add statistical noise
- K-anonymity enforcement

benchmark_metrics:

operational:

- Average wait times
- Appointment utilization
- No-show rates
- Staff productivity

clinical:

- Outcome measures
- Quality scores
- Medication adherence
- Readmission rates

financial:

- Revenue per visit
- Collection rates
- Denial rates
- Cost per procedure

implementation:

data_pipeline:

- Extract from tenant schemas
- Apply privacy transformations
- Aggregate metrics
- Generate benchmarks

access_control:

- Opt-in required
- Role-based visibility
- Audit trail maintained

L.5 Tenant Resource Management

Resource Quotas & Limits

yaml

Tenant_Resource_Quotas:

plan_based_limits:

starter:

users: 10

storage: 10GB

api_calls_per_month: 100000

documents_per_month: 1000

ai_requests_per_month: 500

professional:

users: 50

storage: 100GB

api_calls_per_month: 1000000

documents_per_month: 10000

ai_requests_per_month: 5000

enterprise:

users: Unlimited

storage: 1TB+

api_calls_per_month: Unlimited

documents_per_month: Unlimited

ai_requests_per_month: Unlimited

enforcement:

soft_limits:

notification: 80% threshold

warning: 90% threshold

hard_limits:

action: Reject requests

error_code: QUOTA_EXCEEDED

upgrade_prompt: true

monitoring:

metrics_tracked:

- Current usage by resource
- Trend analysis
- Projection to limit
- Cost implications

dashboards:

- Tenant admin dashboard

- Platform admin dashboard
- Billing dashboard

Appendix M: Business Continuity Planning

M.1 Pandemic/Emergency Response

Remote Operations Procedures

yaml

Emergency_Response_Plan:

activation_triggers:

- Government emergency declaration
- Facility unavailability
- Staff availability < 50%
- Infrastructure compromise

remote_operations:

infrastructure:

- VPN capacity increase to 100%
- Cloud desktop provisioning
- Collaboration tool scaling
- Bandwidth allocation

communication:

primary: Slack/Teams

secondary: Email

emergency: Phone tree

status_page: Public updates

modified_procedures:

- Extended maintenance windows
- Relaxed change approval
- Emergency access protocols
- Simplified documentation

essential_services:

must_maintain:

- Patient portal access
- Clinical documentation
- Prescription management
- Emergency communications

can_defer:

- New feature development
- Non-critical updates
- Training sessions
- Routine maintenance

M.2 Key Person Dependencies

Knowledge Management

yaml

Key_Person_Risk_Mitigation:

critical_roles:

database_administrator:

primary: John Smith

backup: Sarah Johnson

tertiary: External contractor

documentation:

- Database architecture diagram
- Maintenance procedures
- Troubleshooting guide
- Vendor contacts

security_architect:

primary: Alice Chen

backup: Bob Williams

knowledge_transfer:

- Quarterly shadowing
- Documented procedures
- Recorded training sessions
- External audit validation

documentation_requirements:

standard_operating_procedures:

- Step-by-step guides
- Video walkthroughs
- Decision trees
- Contact lists

architecture_documentation:

- System diagrams
- Data flow maps
- Integration points
- Security boundaries

runbooks:

- Incident response
- Disaster recovery
- Routine maintenance
- Emergency procedures

succession_planning:

immediate: Documented handover procedures

short_term: Cross-training program

long_term: Career development paths

M.3 Vendor Failure Scenarios

Vendor Risk Management

yaml

Vendor_Contingency_Plans:

critical_vendors:

cloud_llm_provider:

primary: Google (Gemini)

failover: Vertex AI custom models

contingency: Self-hosted models

switching_procedure:

- Update API endpoints
- Adjust prompt templates
- Validate output quality
- Monitor performance

payment_processor:

primary: Stripe

secondary: Square

manual_fallback: Check processing

communication_services:

sms:

primary: Twilio

backup: AWS SNS

email:

primary: SendGrid

backup: AWS SES

vendor_monitoring:

health_checks:

- API availability
- Response times
- Error rates
- Service status

contract_management:

- SLA tracking
- Renewal dates
- Termination clauses
- Data portability

data_portability:

requirements:

- Standard format exports
- API-based extraction

- Documented schemas
- Migration tools

M.4 Financial Continuity

Cash Flow Management

yaml

Financial_Continuity:

revenue_protection:

diversification:

- Multiple customer segments
- Geographic distribution
- Contract timing variety
- Service line mix

collection_acceleration:

- Automated billing
- Early payment discounts
- Aggressive follow-up
- Payment plans

cost_management:

variable_costs:

- Cloud infrastructure scaling
- Contractor utilization
- Third-party services
- Marketing spend

fixed_cost_reduction:

- Renegotiate contracts
- Defer capital expenses
- Reduce discretionary spending
- Implement hiring freeze

emergency_funding:

sources:

- Line of credit
- Investor bridge loan
- Asset-based lending
- Revenue-based financing

triggers:

- Cash < 90 days runway
- Major customer loss
- Force majeure event
- Regulatory change

N.1 Data Quality Management

Quality Control Framework

yaml

Data_Quality_Framework:

dimensions:

completeness:

definition: Required fields populated

threshold: 95%

validation_rules:

- Patient records: Name, DOB, Gender required
- Encounters: Date, Provider, Diagnosis required
- Medications: Drug, Dose, Frequency required

accuracy:

definition: Data matches source of truth

threshold: 99%

validation_checks:

- Date logic (DOB < Encounter date)
- Code validity (ICD-10, CPT codes)
- Reference integrity

consistency:

definition: Same data across systems

threshold: 98%

reconciliation:

- Cross-system matching
- Duplicate detection
- Format standardization

timeliness:

definition: Data available when needed

sla_targets:

- Real-time: < 1 second
- Near real-time: < 1 minute
- Batch: < 24 hours

quality_monitoring:

automated_checks:

- Completeness scoring
- Anomaly detection
- Trend analysis
- Threshold alerting

quality_dashboards:

metrics:

- Quality score by entity
- Error rates by source
- Remediation progress
- Trend visualization

remediation_workflow:

- Automatic correction where possible
- Queue for manual review
- Root cause analysis
- Process improvement

N.2 Master Data Management

Golden Record Strategy

yaml

Master_Data_Management:

patient_golden_record:

source_systems:

- EHR systems
- Registration systems
- Lab systems
- Billing systems

matching_algorithm:

deterministic_rules:

- Exact SSN match: 100% confidence
- Name + DOB + Address: 95% confidence
- Phone + Email + Name: 85% confidence

probabilistic_matching:

- Levenshtein distance for names
- Soundex for phonetic matching
- Address standardization
- Weighted scoring

survivorship_rules:

priority_order:

- 1: EHR (most trusted)
- 2: Registration
- 3: Billing
- 4: External sources

field_level_rules:

- Most recent for addresses
- Most complete for demographics
- Authoritative source for clinical

merge_procedures:

automatic_merge:

threshold: 95% confidence

manual_review:

threshold: 70-94% confidence

audit_trail:

- Original values preserved
- Merge timestamp

- User/system that merged
- Confidence score

N.3 Data Lineage Tracking

Lineage Documentation

yaml

Data_Lineage:

tracking_implementation:

metadata_capture:

- Source system
- Extraction timestamp
- Transformation applied
- Load timestamp
- Target location

lineage_graph:

nodes:

- Data sources
- Transformations
- Destinations

edges:

- Data flow direction
- Dependencies
- Timing relationships

technical_implementation:

tools:

- Apache Atlas for catalog
- dbt for transformation lineage
- Custom metadata store

api_endpoints:

- GET /lineage/{entity_id}
- GET /impact-analysis/{source_id}
- GET /data-provenance/{record_id}

use_cases:

compliance:

- Demonstrate data sourcing
- Track PHI movement
- Audit data access

operations:

- Impact analysis
- Debugging data issues
- Performance optimization

N.4 Data Catalog

Metadata Management

yaml

Data_Catalog:

catalog_structure:

business_metadata:

- Business description
- Owner/steward
- Criticality level
- Retention policy

technical_metadata:

- Schema definition
- Data types
- Constraints
- Statistics

operational_metadata:

- Update frequency
- Quality metrics
- Usage statistics
- Access patterns

catalog_entries:

datasets:

- Tables
- Views
- Files
- APIs

data_products:

- Reports
- Dashboards
- ML models
- Analytics

search_capabilities:

- Full-text search
- Tag-based filtering
- Schema search
- Lineage exploration

governance_integration:

- Access request workflow
- Data classification

- Compliance tagging
- Quality scoring

N.5 Privacy Impact Assessments

PIA Process

yaml

Privacy_Impact_Assessment:

assessment_triggers:

- New data collection
- System changes
- Third-party sharing
- New use cases

assessment_template:

project_information:

- Project name and description
- Business justification
- Stakeholders involved

data_analysis:

- Data types collected
- Collection methods
- Storage locations
- Retention periods

privacy_risks:

- Unauthorized access
- Data breaches
- Function creep
- Third-party risks

mitigation_measures:

- Technical controls
- Administrative controls
- Physical controls

approval_workflow:

- Privacy officer review
- Security team review
- Legal review
- Executive approval

monitoring:

- Quarterly reviews
- Change-triggered updates
- Annual audits

O.1 Help Desk System Integration

Support Platform Configuration

yaml

Support_System:

platform: Zendesk/Freshdesk

integration_points:

sso_authentication:

protocol: SAML 2.0

identity_provider: Auth0

user_provisioning: SCIM

data_sync:

customer_data:

- Sync from CRM
- Tenant information
- Plan details
- Usage metrics

ticket_integration:

- API webhook on ticket creation
- Status sync bidirectional
- Attachment handling

automation:

ticket_routing:

- Tenant-based assignment
- Priority classification
- Skill-based routing

auto_responses:

- Acknowledgment emails
- FAQ suggestions
- Status updates

O.2 Knowledge Base Architecture

Documentation Portal

yaml

Knowledge_Base:

structure:

public_documentation:

- Getting started guides
- Feature documentation
- Video tutorials
- API reference

authenticated_content:

- Account-specific guides
- Advanced configurations
- Troubleshooting guides
- Best practices

internal_documentation:

- Support runbooks
- Escalation procedures
- Known issues
- Workarounds

content_management:

versioning:

- Documentation versions match product
- Change tracking
- Review workflow

search:

- Full-text search
- Faceted filtering
- AI-powered recommendations
- Usage analytics

delivery:

channels:

- Web portal
- In-app help
- API documentation
- PDF exports

O.3 Support Tier Definitions

Tiered Support Model

yaml

Support_Tiers:

L1_frontline:

responsibilities:

- Password resets
- Basic troubleshooting
- Known issue identification
- Ticket classification

sla:

first_response: 1 hour

resolution_target: 4 hours

escalation_threshold: 2 hours

skills:

- Customer service
- Product knowledge
- Basic technical skills

L2_technical:

responsibilities:

- Configuration issues
- Integration problems
- Performance issues
- Bug triage

sla:

first_response: 2 hours

resolution_target: 24 hours

escalation_threshold: 8 hours

skills:

- Technical troubleshooting
- Log analysis
- SQL queries
- API knowledge

L3_engineering:

responsibilities:

- Code-level issues
- Infrastructure problems
- Complex integrations
- Root cause analysis

sla:

first_response: 4 hours

resolution_target: 72 hours

skills:

- Software development
- System architecture
- Database expertise
- Cloud infrastructure

O.4 SLA Definitions

Service Level Agreements

yaml

SLA_Matrix:

response_times:

critical:

description: Production down, data loss risk

initial_response: 15 minutes

update_frequency: 30 minutes

resolution_target: 4 hours

high:

description: Major feature unavailable

initial_response: 1 hour

update_frequency: 2 hours

resolution_target: 8 hours

medium:

description: Minor feature issue

initial_response: 4 hours

update_frequency: Daily

resolution_target: 48 hours

low:

description: Questions, how-to

initial_response: 8 hours

update_frequency: As needed

resolution_target: 5 days

availability_commitments:

enterprise: 99.95% (22 min/month)

professional: 99.9% (44 min/month)

starter: 99.5% (3.6 hours/month)

credits:

< 99.95%: 10% credit

< 99.9%: 20% credit

< 99.0%: 50% credit

< 95.0%: 100% credit

Appendix P: Legal & Regulatory Framework

P.1 GDPR Compliance

EU Data Protection Requirements

yaml

GDPR_Compliance:

lawful_basis:

processing_grounds:

- Consent (explicit, withdrawable)
- Contract fulfillment
- Legal obligation
- Vital interests
- Legitimate interests

data_subject_rights:

right_to_access:

implementation: Self-service portal

response_time: 30 days

format: Machine-readable

right_to_rectification:

implementation: In-app editing

audit_trail: All changes logged

right_to_erasure:

implementation: Soft delete with purge

exceptions: Legal retention requirements

right_to_portability:

formats: JSON, CSV, FHIR

delivery: Secure download

right_to_object:

opt_out_mechanisms: Preference center

marketing_suppression: Immediate

privacy_by_design:

data_minimization:

- Collect only necessary data
- Automatic data expiry
- Purpose limitation

security_measures:

- Encryption by default
- Pseudonymization
- Access controls
- Regular testing

breach_notification:
authority_notification: 72 hours
individual_notification: Without undue delay
documentation: Breach register maintained

P.2 State-Specific Regulations

State Privacy Laws

yaml

State_Regulations:

california_ccpa:

requirements:

- Privacy policy updates
- Opt-out mechanisms
- Data inventory
- Vendor agreements

consumer_rights:

- Know what data collected
- Delete personal information
- Opt-out of sale
- Non-discrimination

texas_hb300:

requirements:

- Enhanced breach notification
- Training requirements
- Broader PHI definition
- Increased penalties

illinois_bipa:

biometric_data:

- Written consent required
- Retention limitations
- Security requirements
- Private right of action

implementation:

geo_detection: IP-based state identification

consent_management: State-specific flows

policy_versions: Per-state privacy policies

P.3 Business Associate Agreements

BAA Template Structure

yaml

BAA_Template:

standard_provisions:

permitted_uses:

- Treatment activities
- Payment operations
- Healthcare operations
- As required by law

required_safeguards:

- Administrative safeguards
- Physical safeguards
- Technical safeguards
- Organizational requirements

breach_obligations:

- Immediate notification
- Mitigation efforts
- Documentation provision
- Cooperation with investigation

termination:

- Material breach cure period
- Immediate for uncured breach
- Data return/destruction
- Survival of obligations

subcontractor_requirements:

flow_down_provisions:

- Same restrictions apply
- Written agreement required
- Compliance monitoring

liability:

indemnification: Mutual

insurance_requirements:

- **Cyber liability:** \$5M minimum
- **Professional liability:** \$2M minimum
- **General liability:** \$2M minimum

P.4 Data Processing Agreements

DPA Requirements

yaml

Data_Processing_Agreement:

processor_obligations:

processing_instructions:

- Written instructions only
- Purpose limitation
- Confidentiality binding

security_measures:

- Industry standard security
- Regular security updates
- Incident response procedures

sub_processors:

- Prior written consent
- Updated list maintained
- Liability for sub-processors

audit_rights:

- Annual audits permitted
- Certification acceptable
- Cost allocation defined

data_location:

- Specified regions only
- Transfer mechanisms defined
- Restrictions documented

controller_obligations:

lawful_instructions:

- Compliance with laws
- Necessary authorizations
- Accuracy of instructions

cooperation:

- Data subject requests
- Regulatory inquiries
- Security assessments

P.5 Insurance Requirements

Cyber Insurance Coverage

yaml

Insurance_Requirements:

cyber_liability:

minimum_coverage: \$10M

covered_events:

- Data breaches
- Ransomware attacks
- Business interruption
- Regulatory fines
- Notification costs
- Credit monitoring
- Legal defense

professional_liability:

minimum_coverage: \$5M

covered_events:

- Service failures
- Professional negligence
- Misrepresentation
- IP infringement

general_liability:

minimum_coverage: \$2M

covered_events:

- Bodily injury
- Property damage
- Personal injury
- Advertising injury

additional_coverage:

directors_officers: \$5M

employment_practices: \$2M

crime_coverage: \$1M

Appendix Q: Financial Operations (FinOps)

Q.1 Usage Metering Architecture

Resource Tracking System

yaml

Usage_Metering:

metrics_collected:

compute:

- API calls by endpoint
- Processing time
- CPU/memory usage
- Concurrent users

storage:

- Data stored (GB)
- Bandwidth used
- File operations
- Backup storage

ai_services:

- LLM tokens consumed
- Document pages processed
- Voice minutes transcribed
- ML predictions made

collection_pipeline:

instrumentation:

- Application-level metrics
- Infrastructure metrics
- Database metrics
- Third-party usage

aggregation:

- Per-minute raw data
- Hourly aggregates
- Daily summaries
- Monthly billing data

storage:

raw_metrics: Cloud Monitoring (30 days)

aggregates: BigQuery (7 years)

billing_data: Cloud Storage (10 years)

Q.2 Billing Integration

Payment Processing Setup

yaml

Billing_System:

stripe_integration:

configuration:

api_version: "2024-01-01"

webhook_endpoint: /api/webhooks/stripe

subscription_management:

plans:

- starter_monthly
- professional_monthly
- enterprise_annual

features:

- Usage-based billing
- Metered billing
- Tiered pricing
- Volume discounts

payment_methods:

- Credit/debit cards
- ACH transfers
- Wire transfers
- Invoicing

quickbooks_integration:

sync_frequency: Daily

data_mapping:

- Customers → QB Customers
- Invoices → QB Invoices
- Payments → QB Payments
- Credits → QB Credit Memos

reconciliation:

- Daily balance checks
- Monthly statements
- Audit trail maintenance

Q.3 Cost Attribution

Multi-Tenant Cost Allocation

yaml

Cost_Attribution:

allocation_methods:

direct_costs:

- Dedicated resources
- Third-party API usage
- Storage consumption
- Support tickets

shared_costs:

allocation_basis:

- Usage percentage
- User count
- Transaction volume
- Revenue share

cost_models:

infrastructure:

compute:

formula: $(\text{CPU_hours} * \text{rate}) + (\text{Memory_GB_hours} * \text{rate})$

storage:

formula: $(\text{GB_stored} * \text{days} * \text{rate}) + (\text{bandwidth_GB} * \text{rate})$

database:

formula: $\text{Base_cost} * (\text{tenant_usage} / \text{total_usage})$

per_unit_costs:

api_call: \$0.0001

document_page: \$0.01

ai_request: \$0.05

storage_gb_month: \$0.10

reporting:

tenant_invoice:

- Line items by service
- Usage details
- Cost breakdown
- Month-over-month comparison

internal_reporting:

- Cost per tenant
- Margin analysis

- Resource efficiency
- Forecasting

Q.4 Budget Management

Cost Control Framework

yaml

Budget_Controls:

budget_hierarchy:

organization:

annual_budget: \$2M

quarterly_allocation: \$500K

department:

engineering: 40%

infrastructure: 30%

third_party: 20%

other: 10%

project:

tracked_separately: true

approval_required: >\$10K

alerts:

thresholds:

- 50%: Email notification
- 75%: Slack alert
- 90%: Manager escalation
- 100%: Spending freeze

forecasting:

- Linear projection
- Seasonal adjustment
- Growth factoring

optimization:

automated:

- Idle resource termination
- Reserved instance purchasing
- Spot instance usage
- Storage lifecycle policies

manual_review:

frequency: Monthly

focus_areas:

- Unused resources
- Oversized instances
- Data transfer costs
- License optimization

Appendix R: Training & Change Management

R.1 User Training Programs

Role-Based Training Curricula

yaml

Training_Programs:

clinical_users:

modules:

basic:

- System navigation
- Patient search
- Documentation basics
- Safety features

duration: 2 hours

advanced:

- Clinical workflows
- Order management
- Result review
- Reporting

duration: 4 hours

specialty:

- Department-specific workflows
- Custom templates
- Integration points

duration: 2 hours

delivery_methods:

- Interactive webinars
- Self-paced modules
- In-person workshops
- Simulation environment

administrative_users:

modules:

system_configuration:

- User management
- Role configuration
- Workflow setup
- Template management

duration: 6 hours

reporting:

- Report generation
- Dashboard creation
- Data exports
- Analytics tools

duration: 4 hours

technical_users:

modules:

api_integration:

- Authentication
- API endpoints
- Webhooks
- Rate limiting

duration: 8 hours

administration:

- System monitoring
- Troubleshooting
- Performance tuning
- Security management

duration: 8 hours

R.2 Administrator Guides

System Administration Documentation

yaml

Admin_Documentation:

configuration_guide:

sections:

- Initial setup
- Organization structure
- User provisioning
- Security settings
- Integration configuration
- Customization options

operational_procedures:

daily_tasks:

- System health checks
- User access reviews
- Backup verification
- Alert monitoring

weekly_tasks:

- Performance review
- Security updates
- Usage analytics
- Capacity planning

monthly_tasks:

- Compliance audit
- Disaster recovery test
- Documentation update
- Training review

troubleshooting_guide:

common_issues:

- Login problems
- Performance issues
- Integration failures
- Data discrepancies

diagnostic_tools:

- Log analysis
- System monitors
- Debug modes
- Support tools

R.3 Developer Resources

API Documentation & SDKs

yaml

Developer_Resources:

api_documentation:

format: OpenAPI 3.0

sections:

- Authentication guide
- Endpoint reference
- Data models
- Code examples
- Rate limits
- Error handling

sdk:

languages:

javascript:

package: "@sarathi/sdk-js"

features:

- TypeScript support
- Promise-based
- Automatic retry
- Request batching

python:

package: "sarathi-sdk"

features:

- Async support
- Type hints
- Pagination handling
- Error recovery

java:

package: "com.sarathi.sdk"

features:

- Spring integration
- Reactive streams
- Circuit breaker
- Connection pooling

code_examples:

- Authentication flows
- CRUD operations
- Bulk operations
- Webhook handling

- Error scenarios
- Best practices

developer_portal:

features:

- Interactive API explorer
- Sandbox environment
- Code generation
- Testing tools
- Community forum

R.4 Change Advisory Board

CAB Procedures

yaml

Change_Advisory_Board:

composition:

permanent_members:

- CTO
- VP Engineering
- Security Officer
- Operations Manager

rotating_members:

- Product Manager
- Customer Representative
- Technical Lead

change_categories:

standard:

approval: Pre-approved

examples:

- Security patches
- Minor updates
- Documentation changes

normal:

approval: CAB review

lead_time: 5 days

examples:

- Feature releases
- Configuration changes
- Integration updates

emergency:

approval: Expedited

lead_time: Immediate

examples:

- Security fixes
- Critical bugs
- Outage resolution

review_process:

submission:

- Change request form
- Impact assessment
- Rollback plan
- Testing evidence

evaluation:

- Risk assessment
- Resource availability
- Conflict checking
- Approval decision

post_implementation:

- Success verification
- Lesson learned
- Documentation update

R.5 Feature Adoption Tracking

Usage Analytics Framework

yaml

Adoption_Tracking:

metrics:

feature_usage:

- Active users per feature
- Usage frequency
- Time to first use
- Feature retention

user_journey:

- Onboarding completion
- Feature discovery
- Task completion rates
- User paths

satisfaction:

- In-app surveys
- NPS scores
- Support tickets
- Feature requests

analysis_tools:

product_analytics:

platform: Mixpanel/Amplitude

events_tracked:

- Feature interactions
- Workflow completion
- Error encounters
- Performance metrics

user_feedback:

channels:

- In-app feedback
- Email surveys
- User interviews
- Focus groups

reporting:

dashboards:

- Feature adoption dashboard
- User engagement metrics
- Retention analysis
- Cohort analysis

stakeholder_reports:

- Monthly adoption report
- Quarterly business review
- Annual platform assessment

Appendix S: Analytics Platform Architecture

S.1 Data Warehouse Schema

Dimensional Model Design

yaml

Data_Warehouse_Schema:

fact_tables:

fact_encounters:

grain: One row per patient encounter

dimensions:

- dim_date
- dim_patient
- dim_provider
- dim_facility
- dim_diagnosis
- dim_procedure

measures:

- encounter_duration
- total_charges
- total_payments
- procedure_count
- diagnosis_count

fact_appointments:

grain: One row per appointment

dimensions:

- dim_date
- dim_time
- dim_patient
- dim_provider
- dim_appointment_type

measures:

- scheduled_duration
- actual_duration
- wait_time
- no_show_flag

fact_clinical_metrics:

grain: One row per metric per patient per date

dimensions:

- dim_date
- dim_patient
- dim_metric_type

- dim_provider

measures:

- metric_value
- normal_range_flag
- change_from_previous

dimension_tables:

dim_patient:

attributes:

- patient_key (surrogate)
- patient_id (natural)
- demographics
- address
- insurance
- risk_factors

slowly_changing:

type: SCD Type 2

tracked_changes:

- Address
- Insurance
- Risk factors

dim_provider:

attributes:

- provider_key
- provider_id
- name
- specialty
- department
- credentials

dim_date:

attributes:

- date_key
- full_date
- year
- quarter
- month
- week
- day_of_week

- is_holiday
- is_weekend

S.2 Healthcare KPIs & Operational Metrics

Healthcare Metrics

yaml

Key_Performance_Indicators:

financial_kpis:

total_revenue:

definition: Total practice revenue

calculation: SUM(payments + adjustments)

frequency: Daily

target: \$325,000/month

trend_tracking: +5.8% MoM

collections:

definition: Actual payments received

calculation: SUM(patient_payments + insurance_payments)

frequency: Daily

target: \$280,000/month

trend_tracking: +3.2% MoM

accounts_receivable:

definition: Outstanding balances

calculation: SUM(unbilled + unpaid_claims)

frequency: Daily

target: <\$500,000

trend_tracking: -0.5% MoM

days_in_ar:

definition: Average collection time

calculation: accounts_receivable / daily_charges

frequency: Daily

target: <42 days

trend_tracking: +\$12 MoM

payer_profitability:

definition: Revenue per payer per encounter

ranking_metrics:

- Average reimbursement rate
- Denial rate by payer
- Time to payment

intelligence_alerts:

- Below-market reimbursement detection
- High denial rate patterns
- Contract negotiation opportunities

operational_kpis:

patient_volume:

definition: Unique patients seen

calculation: $\text{COUNT}(\text{DISTINCT patient_id})$

frequency: Daily

target: 450 per day

trend_tracking: +12% MoM

appointment_utilization:

definition: Filled vs available slots

calculation: $\text{filled_slots} / \text{total_slots} * 100$

frequency: Daily

target: 85%

current: 87%

no_show_rate:

definition: Missed appointments

calculation: $\text{no_shows} / \text{scheduled} * 100$

frequency: Daily

target: <8%

trend_tracking: +2% MoM

ai_insights:

- 30% higher for appointments >2 months out
- Monday mornings 25% higher wait times
- Dr. Adams 15% lower than average

average_wait_time:

definition: Check-in to seen time

calculation: $\text{AVG}(\text{seen_time} - \text{checkin_time})$

frequency: Real-time

target: <20 minutes

trend_tracking: -15ms MoM

new_patient_percentage:

definition: New vs returning patients

calculation: $\text{new_patients} / \text{total_patients} * 100$

frequency: Weekly

target: 15%

trend_tracking: +5 patients MoM

clinical_kpis:

care_gap_closure:

definition: Preventive care completion

calculation: $\text{completed_measures} / \text{due_measures} * 100$

frequency: Monthly

target: >75%

medication_adherence:

definition: Prescriptions filled on time

calculation: $\text{on_time_fills} / \text{total_prescriptions} * 100$

frequency: Monthly

target: >80%

lab_turnaround_time:

definition: Order to result time

calculation: $\text{AVG}(\text{result_time} - \text{order_time})$

frequency: Daily

target: <48 hours

provider_kpis:

provider_productivity:

metrics:

- Patients per day
- RVUs generated
- Documentation time
- Patient satisfaction scores

comparison: Side-by-side provider metrics

referral_metrics:

top_referrers:

- Volume by referring physician
- Conversion rate by source
- Leakage detection (15% to competitors)

referral_quality:

- Average patient value by referrer
- No-show rate by referral source
- Time to appointment by urgency

ai_agent_kpis:

agent_utilization:

definition: AI task completion rate

calculation: $\text{successful_completions} / \text{total_attempts} * 100$

target: >90%

current: 92%

automation_impact:

time_saved: 450 hours/month

tasks_automated: 12,500/month

cost_reduction: \$45,000/month

agent_performance:

response_times:

- Document processing: 1500ms avg
- Clinical notes: 450ms avg
- Patient messaging: 120ms avg

accuracy_rates:

- OCR extraction: 95%
- Code suggestion: 89%
- Prior auth approval: 78%

scaling_insights:

- Monday peak loads identified
- Morning rounds 15% higher usage
- Claim processing 68% faster

revenue_intelligence:

ai_forecast:

method: Historical trend + scheduled appointments

accuracy: 92% within 5%

positive_drivers:

- 15% increase in high-value procedures
- Seasonal volume increases

negative_drivers:

- Provider vacation impacts
- Payer mix changes

negotiation_intelligence:

opportunities:

- CPT codes below regional average
- High-volume procedures for leverage
- Denial pattern improvements

alerts:

- Contract renewal dates
- Rate change notifications
- New payer opportunities

S.3 Dashboard Templates

Pre-Built Dashboards

yaml

Dashboard_Library:

ai_agents_dashboard:

url: /ai-agents

purpose: Real-time monitoring of AI agents and workflows

widgets:

- Agent health status indicators
- Daily/monthly/annual invocation counts
- Automation success rates
- Background agent task tracking
- User-in-loop feature metrics

key_metrics:

- 12,500 user tasks enhanced by AI monthly
- 92% automation success rate
- 450 hours saved per month
- Agent-specific performance tracking

refresh: Real-time with 60-second polling

practice_command_center:

url: /practice-owner-portal

purpose: 360-degree view of practice financial and operational health

widgets:

- Revenue and collections trends
- Days in AR tracking
- Patient volume metrics
- Provider productivity comparison
- AI agent usage patterns
- Referral source analytics
- Payer profitability rankings

intelligence_features:

- AI-driven revenue forecasting
- Positive/negative revenue drivers
- Referral leakage detection (15%)
- Payer negotiation insights

refresh: Real-time with daily aggregations

practice_health_dashboard:

url: /practice-health

purpose: Monitor service status and performance

widgets:

- Internal service health (AI Scribe, Portal, Billing)
- External integration status (Labs, Pharmacies, Insurance)
- Response time monitoring
- Service dependency mapping
- Alert status indicators

monitoring:

- Latency tracking (target vs actual)
- Uptime percentages
- Error rate monitoring
- API rate limit usage

alert_thresholds:

- >500ms latency: Yellow
- >2000ms latency: Red
- **Service down:** Critical alert

referral_management_portal:

url: /referral-portal

purpose: Streamline and track incoming referrals

widgets:

- Referral intake queue
- AI urgency classification
- Status tracking workflow
- Referring physician directory
- Conversion analytics

features:

- Automated urgency assessment
- One-click scheduling
- Referral source tracking
- Outcome reporting

workflow_states:

- Pending Review
- Contacted
- Scheduled
- Completed

executive_dashboard:

widgets:

- Monthly revenue trends with forecast
- Key operational metrics summary
- Provider performance rankings
- Quality measure compliance
- AI automation ROI tracking

filters:

- Date range selector
- Facility filter
- Department breakdown
- Provider comparison

clinical_dashboard:

widgets:

- Patient panel management
- Care gap identification
- Pending lab results
- Medication alerts
- Clinical protocol reminders

personalization:

- Provider-specific views
- Specialty customization
- Custom alert preferences

operational_dashboard:

widgets:

- Real-time appointment schedule
- Room utilization tracking
- Staff coverage matrix
- Wait time monitoring
- Workflow bottleneck detection

drill_down:

- Hourly utilization details
- Provider-level metrics
- Department comparisons

financial_dashboard:

widgets:

- Revenue cycle metrics
- Payer mix analysis
- Denial trending

- Collection rate tracking
- Cost per visit analysis

comparisons:

- Period-over-period trending
- Budget vs actual variance
- Industry benchmark comparison

S.4 Regulatory Reporting

Compliance Reports

yaml

Regulatory_Reports:

quality_reporting:

mips:

measures:

- Quality measures
- Cost measures
- Improvement activities
- Promoting interoperability

submission:

frequency: Annual

format: QRDA III

deadline: March 31

clinical_quality_measures:

specifications:

- Diabetes care
- Hypertension control
- Preventive care
- Depression screening

calculation:

numerator: Patients meeting criteria

denominator: Eligible patients

exclusions: Documented exceptions

meaningful_use:

attestation_requirements:

- Electronic prescribing
- Clinical decision support
- Patient portal usage
- Health information exchange

reporting_period: 90 days minimum

public_health_reporting:

immunization_registry:

frequency: Real-time

format: HL7 2.5.1

syndromic_surveillance:

frequency: Daily

format: HL7 CDA

reportable_conditions:

trigger: Automatic on diagnosis

submission: Within 24 hours

S.5 Predictive Analytics

ML Model Deployment

yaml

Predictive_Models:

risk_stratification:

model_type: Gradient boosting

features:

- Demographics
- Diagnoses
- Medications
- Lab results
- Utilization history

predictions:

- Readmission risk
- ED visit probability
- Medication non-adherence
- Care gap risk

deployment:

platform: Vertex AI

update_frequency: Monthly

monitoring: Drift detection

appointment_no_show:

model_type: Logistic regression

features:

- Historical no-shows
- Appointment type
- Day of week
- Weather data
- Transportation access

interventions:

high_risk:

- Double confirmation
- Transportation assistance
- Overbook consideration

revenue_forecasting:

model_type: Time series (Prophet)

inputs:

- Historical revenue

- Appointment schedule
- Payer mix
- Seasonal patterns

outputs:

- 30-day forecast
- 90-day forecast
- Confidence intervals

Appendix T: Observability & SRE Practices

T.1 Service Level Objectives

SLO Definitions

```
yaml
```

Service_Level_Objectives:

api_availability:

sli: Successful requests / Total requests

slo_target: 99.95%

measurement_window: 30 days

error_budget:

monthly_budget: 21.6 minutes

burn_rate_alerts:

- 1_hour: >14.4% budget burned
- 6_hours: >10% budget burned
- 24_hours: >5% budget burned

request_latency:

sli: Requests < 200ms / Total requests

slo_target: 95%

percentile_targets:

p50: <50ms

p95: <200ms

p99: <500ms

data_durability:

sli: Successful data retrievals / Total retrievals

slo_target: 99.999%

backup_validation:

frequency: Daily

restoration_test: Weekly

clinical_workflows:

prescription_processing:

sli: Prescriptions processed < 30s

slo_target: 99%

lab_result_delivery:

sli: Results available < 5 minutes

slo_target: 95%

T.2 Error Budget Management

Error Budget Policy

yaml

Error_Budget_Management:

budget_calculation:

formula: $1 - \text{SLO_target}$

example: $1 - 0.9995 = 0.05\% = 21.6 \text{ min/month}$

budget_consumption_triggers:

25_percent:

actions:

- Review recent changes
- Increase monitoring

50_percent:

actions:

- Reduce deployment velocity
- Focus on reliability

75_percent:

actions:

- Freeze non-critical changes
- All hands on reliability

100_percent:

actions:

- Stop all feature work
- Emergency reliability sprint

budget_recovery:

conditions:

- Root cause identified
- Fixes implemented
- Monitoring enhanced
- Runbooks updated

T.3 Chaos Engineering

Failure Injection Framework

yaml

Chaos_Engineering:

experiment_types:

infrastructure:

- Random pod termination
- Network latency injection
- Disk space exhaustion
- CPU throttling

application:

- Database connection drops
- Cache failures
- API timeouts
- Message queue delays

dependencies:

- Third-party API failures
- Authentication service outage
- Payment gateway issues

experiment_schedule:

development:

frequency: Daily

scope: Component level

staging:

frequency: Weekly

scope: Service level

production:

frequency: Monthly

scope: Controlled, single service

blast_radius: <5% traffic

safety_mechanisms:

- Automatic rollback on SLO breach
- Manual abort button
- Exclude critical periods
- Customer notification (if needed)

T.4 Runbook Automation

Self-Healing Procedures

yaml

Automated_Runbooks:

common_issues:

high_memory_usage:

trigger: Memory > 90% for 5 minutes

automated_actions:

- Clear application cache
- Force garbage collection
- Scale horizontally
- Alert if persists

database_connection_pool:

trigger: Available connections < 10

automated_actions:

- Kill idle connections
- Increase pool size
- Restart connection pool
- Page on-call if